Extending Partition-Theoretic Prime Detection: A Computational Study of the Weight-12 Barrier, the Expression E_5, and a Basis-Dependent Recovery of E_6

Nigel Randsley

Abstract

Craig, van Ittersum, and Ono [1] introduced non-negative polynomial combinations of MacMahon partition functions that vanish precisely at the primes, producing explicit expressions $E_1, \ldots, E_4$ and motivating a broader finite-generation picture. This paper presents a computational extension of that framework in exact rational arithmetic. Four bounded results are established. First, within the tested prime-evaluation matrices built from the basis $\{M_1, \ldots, M_6\}$, the columns attached to M_6 are pivot columns for all tested degrees, so no new prime-vanishing direction involving M_6 appears in that restricted basis. Second, after restricting to $\{M_1, \ldots, M_5\}$, one finds a new primitive prime-vanishing direction E_5 appearing already at polynomial degree 2. Third, after enlarging the basis to $\{M_1, \ldots, M_7\}$, a further new direction E_6 appears at degree 4, consistent with a cancellation of cusp-form contributions involving Ramanujan's τ-function. Fourth, exhaustive exact searches through $a_{\max} = 12$ and degree $d \leq 8$ reveal no additional independent directions. The paper is framed explicitly as a computational study: finite verified claims are separated from modular-form interpretation, the distinction between prime-vanishing generators and non-negative prime-detecting expressions is made explicit, and all completeness statements are confined to the tested search range.

1 Introduction

The MacMahon partition functions $M_a(n)$ lie at the intersection of partition theory, divisor-sum identities, and the theory of quasimodular forms. Recent work of Craig, van Ittersum, and Ono [1] showed that suitable non-negative polynomial combinations of these functions vanish at the primes and only at the primes, yielding explicit expressions E_1, E_2, E_3, and E_4. Each E_k in that sequence introduces M_{k+1} as its highest-weight component at an increasing polynomial degree. That result suggests that prime detection by partition-theoretic data may admit a finite algebraic description.

The present paper studies the next stage of that program by exact computation. Its starting point is a structural failure of the naïve continuation pattern. After the first four expressions, the most obvious next step would be to introduce M_6 and continue the degree pattern visible in $E_1, \ldots, E_4$. The computation shows that this does not occur. Instead, the obstruction appears exactly at weight 12, the first weight at which cusp forms enter the relevant modular-form space. This coincidence motivates the central thesis of the paper: the post-E_4 behavior is controlled not by a routine continuation of the low-weight pattern, but by a genuine structural change in the prime-evaluation linear algebra.

We then show that the obstruction is basis-dependent. When the basis is enlarged to $\{M_1, \ldots, M_7\}$, a new cancellation channel opens: a linear combination of M_6 and M_7 can simultaneously annihilate both τ-type contributions at primes, yielding the sixth expression E_6. An exhaustive search through $a_{\max} = 12$ and $d \leq 8$ finds no further independent directions, even at weight 24 where the cusp-form space first becomes two-dimensional.

Related and concurrent work has extended the CIO framework in complementary directions: Kang, Matsusaka, and Shin [2] establish quasi-modularity for level-N MacMahon functions and derive new prime-detecting expressions; Gomez [3] shows that MacMahonesque functions can detect cubes of primes and primes in arithmetic progressions; and van Ittersum, Mauth, Ono, and Singh [4], together with Kane, Krishnamoorthy, and Lau [5], independently proved the CIO conjecture that every level-1 prime-detecting quasimodular form is an Eisenstein series.

The scope of the present paper is deliberately narrow. It does not claim a general theorem beyond what is computationally verified. Its purpose is to isolate, document, and interpret the first post-E_4 phenomena in exact arithmetic. The four main results, each bounded to the tested range, are:

(1) Weight-12 barrier (Section 3). For all tested degrees $d \in \{0, \ldots, 6\}$, the columns of the prime evaluation matrix attached to M_6 are pivot columns, so no prime-vanishing expression in the $\{M_1, \ldots, M_6\}$ basis at any tested degree involves M_6.

(2) Emergence of E_5 (Sections 4–6). Restricting to $\{M_1, \ldots, M_5\}$, a new prime-vanishing direction E_5 first appears at polynomial degree 2; its explicit formula is derived and verified.

(3) Basis-dependent recovery via E_6 (Section 7). Enlarging the basis to $\{M_1, \ldots, M_7\}$, a τ-cancellation mechanism yields a further direction E_6 at degree 4, involving both M_6 and M_7.

(4) Bounded completeness (Section 8). Exhaustive searches through $a_{\max} = 12$ and $d \leq 8$ find no independent directions beyond $E_1, \ldots, E_6$.

Relation to companion papers

The present paper is the first of three works that together form a complete study of prime-vanishing directions in the MacMahon basis. Their roles are distinct and sequential.

Paper I (this paper). *Computational discovery.* Exact rational computation establishes the existence and explicit form of E_5 and E_6, locates the weight-12 cusp-form barrier as the mechanism behind the post-E_4 break in the pattern, and documents the τ-cancellation mechanism that restores a prime-vanishing direction when M_7 is adjoined. All results are stated as bounded, computationally verified claims. No structural proof of the barrier or the recovery is attempted here.

Paper II. *Structural proof of the obstruction.* A companion paper (*A structural obstruction to prime-vanishing MacMahon expressions beyond degree four*) proves unconditionally, via quasimodular form theory, that no prime-vanishing polynomial expression in $\{M_1, \ldots, M_6\}$ exists at any polynomial degree. The argument identifies a one-dimensional *obstruction space* $\mathcal{O}$ generated by the $[\tau(p)]$ class, which is orthogonal to every Eisenstein-type combination of $M_1, \ldots, M_5$ yet cannot be cancelled within the M_6 column alone. The weight-12 barrier discovered computationally in Paper I is thereby given a complete structural explanation.

Paper III. *Structural and computational proof of the first recovery.* A second companion paper (*Recovery of a prime-vanishing direction via M_6 and M_7: structural proof and obstruction dimension*) proves that adjoining M_7 resolves the obstruction. The key steps are a structural identity $D(\Delta) = E_2 \cdot \Delta$ (derived from Ramanujan's differential equations) that fixes the precise cusp content of M_7 at primes, and a full analysis of the resulting $(d{+}2)$-dimensional obstruction space $\mathcal{O}_d$. Together, Papers II and III establish that E_6 is not an accidental computational artefact but the unique first recovery forced by the modular arithmetic of the MacMahon generating functions.

The three papers are designed to be read independently: each contains its own precise statements and proofs. Papers II and III cite the computational discoveries of Paper I for motivation but depend on them only informally.

Position within the broader program. The present paper should be read as the computational discovery paper in a broader three-part program. Its role is to identify, by exact rational search, the first post-E_4 phenomena: the weight-12 obstruction in the restricted $\{M_1, \ldots, M_6\}$ basis, the unexpected appearance of E_5 in $\{M_1, \ldots, M_5\}$, and the first recovery E_6 after adjoining M_7. Subsequent companion work proves the weight-12 obstruction structurally and then analyses the first recovery mechanism through a combination of structural arguments and exact computational verification. In this sense, the present paper establishes the discovery layer, while the companion papers supply the first theorem-level explanation of the obstruction–recovery pattern.

2 Background and Notation

For $a \geq 1$, define the MacMahon partition function by

$$M_a(n) = \sum_{\substack{0<s_1<\cdots<s_a \\ m_i \geq 1,\, \sum m_i s_i = n}} m_1 m_2 \cdots m_a, \tag{1}$$

following the partition-analysis framework of MacMahon [7]. Let $U_a(q) = \sum_{n\geq 1} M_a(n) q^n$ denote the associated generating series. As established in [1], $U_a(q)$ is a quasimodular form of weight $2a$ on $\mathrm{SL}_2(\mathbb{Z})$ in the sense of Kaneko–Zagier [12]; the ring of quasimodular forms on $\mathrm{SL}_2(\mathbb{Z})$ is $\mathbb{Q}[E_2, E_4, E_6]$ [11].

Divisor sums and the low-weight case. For $a \in \{1, 2, 3, 4, 5\}$ (weights $2a \leq 10$), the relevant quasimodular spaces are spanned entirely by Eisenstein series, so $M_a(n)$ can be written as an explicit polynomial combination of divisor power sums $\sigma_k(n) = \sum_{d|n} d^k$. In particular, at any prime p one has $\sigma_k(p) = p^k + 1$, so evaluating $M_a(p)$ yields a polynomial in p; this algebraic collapse creates the linear dependencies that underlie the null space containing $E_1, \ldots, E_4$.

Prime evaluation matrix. Fix a degree $d \geq 0$ and a maximum index $a_{\max} \geq 1$. The *prime evaluation matrix* $\mathbf{M}(d, a_{\max}, N)$ is the $N \times (d+1)a_{\max}$ matrix whose rows are evaluations of $\{n^k M_a(n) : 0 \leq k \leq d,\ 1 \leq a \leq a_{\max}\}$ at the first N primes. All arithmetic is performed exactly over $\mathbb{Q}$. A vector in the null space of $\mathbf{M}(d, a_{\max}, N)$ corresponds to a degree-d polynomial combination of $M_1, \ldots, M_{a_{\max}}$ that vanishes at all N tested primes. We say such a combination is *prime-vanishing*.

Three levels of structure. The paper operates at three distinct levels, summarised in Table 1.

Table 1: Three levels of structure in the present paper.

Object	Meaning	Role in this paper
Prime-vanishing generator	Vanishes at the tested primes	Null-space direction, e.g. E_5
Non-negative prime-detecting expression	Vanishes only at primes and is non-negative on tested composites	Stronger detection object, recovered from generators by adjustment
Structural obstruction/recovery phenomenon	Modular explanation of the disappearance or reappearance of directions	Interprets the weight-12 barrier and the M_7-recovery

3 The Weight-12 Barrier

3.1 Quasimodular forms at weight 12 and Ramanujan's Δ

The space of modular forms of weight 12 on $\mathrm{SL}_2(\mathbb{Z})$ is two-dimensional, spanned by the Eisenstein series E_{12} and the unique normalized cusp form

$$\Delta(q) = q \prod_{k=1}^{\infty} (1 - q^k)^{24} = \sum_{n=1}^{\infty} \tau(n)\, q^n,$$

where $\tau(n)$ is the Ramanujan tau function [8]. Because $U_6(q)$ projects non-trivially onto this cusp-form space, the closed-form of $M_6(n)$ must involve Ramanujan's τ-function. Fitting the 55×22 rational linear system (21 polynomial-weighted divisor-sum columns plus one τ-column) yields

$$M_6(n) = \sum_{j=0}^{5} P_j(n)\, \sigma_{2j+1}(n) - \frac{17}{150\,450\,048\,000}\, \tau(n), \tag{2}$$

where the polynomials $P_j(n)$ are given explicitly in Appendix A. The non-zero coefficient $c_\tau = -17/150\,450\,048\,000$ is the key obstruction; its denominator factors as $2^{10} \times 3^5 \times 5^3 \times 7 \times 691$, where $691 = \mathrm{num}(B_{12})$ is the Bernoulli prime that indexes the weight-12 Eisenstein series and reflects the congruence $\tau(n) \equiv \sigma_{11}(n) \pmod{691}$ first observed by Ramanujan [8] and explained structurally by Swinnerton-Dyer [9] via the congruence of Hecke eigenforms modulo 691.

3.2 Exclusion of M_6 from prime-vanishing expressions

Unlike the divisor sums, $\tau(p)$ does not reduce to a polynomial in p; Deligne's proof of the Weil conjectures [10] gives $|\tau(p)| \le 2p^{11/2}$. Consequently, the τ-component of M_6 contributes data at primes that is algebraically independent of any combination drawn from $\{M_1, \ldots, M_5\}$.

Proposition 3.1 (Computational exclusion of M_6)**.** *For each tested degree $d \in \{0, 1, 2, 3, 4, 5, 6\}$ and $N = 95$ primes in $[2, 500]$,*

$$\mathrm{rank}\ \mathbf{M}(d, 6, 95) = \mathrm{rank}\ \mathbf{M}(d, 5, 95) + (d + 1),$$

and therefore $\dim \mathrm{Ker}\ \mathbf{M}(d, 6, 95) = \dim \mathrm{Ker}\ \mathbf{M}(d, 5, 95)$.

Proof. Explicit computation for $d \in \{2, 3\}$ is shown in Table 2; the identity was verified computationally for all tested d. At $d = 2$: the matrix $\mathbf{M}(2, 5, 95)$ has size 95×15, rank 11, nullity 4. Adding the three columns $\{M_6, nM_6, n^2M_6\}$ gives $\mathbf{M}(2, 6, 95)$ of size 95×18, rank 14, nullity 4. In the rational RREF of $\mathbf{M}(2, 6, 95)$, all three M_6-associated columns are pivot columns, so the null space gains no new direction. □

Table 2: Prime-evaluation matrix dimensions at $d = 2$, illustrating the pivot status of M_6.

Matrix	Rows	Cols	Rank	Nullity
$a_{\max} = 5,\ d = 2$	95	15	11	4
$a_{\max} = 6,\ d = 2$	95	18	14	4

Corollary 3.2. *For each tested degree $d \in \{0, \ldots, 6\}$, no prime-vanishing expression formed from the basis $\{M_1, \ldots, M_6\}$ at degree at most d can have a non-zero M_6 coefficient. Within this basis and degree range, the prime-vanishing null space is identical to that of $\{M_1, \ldots, M_5\}$.*

Remark 3.3. *Corollary 3.2 is basis-specific. When M_7 is adjoined, the combined basis $\{M_1, \ldots, M_7\}$ admits new prime-vanishing directions involving both M_6 and M_7; the expression E_6 is the first such direction (Section 7).*

4 Search Strategy in the Restricted Basis $\{M_1, \ldots, M_5\}$

Having established that M_6 cannot appear in any prime-vanishing expression within the $\{M_1, \ldots, M_6\}$ basis (Corollary 3.2), we restrict attention to $\mathcal{B}_d = \{n^k M_a(n) : 0 \le k \le d,\ 1 \le a \le 5\}$ and sweep over increasing d.

For each d, we compute the rational null space of $\mathbf{M}(d, 5, N)$ and test each null vector for independence from the $\mathbb{Q}[n]$-span of $E_1, \ldots, E_4$: specifically, we evaluate each candidate at a set of composites in $[4, 100]$ and check whether the resulting vector lies outside the column span of $\{E_i(\text{composites})\}_{i=1}^4$ via a rank test over $\mathbb{Q}$. The first degree at which an independent new direction appears is taken as the canonical minimal-degree form of E_5.

5 Emergence and Formula of E_5

Proposition 5.1 (First new direction at degree 2)**.** *In the exact search over the basis $\{M_1, \ldots, M_5\}$ with $N = 95$ primes, the null space of $\mathbf{M}(d, 5, 95)$ is trivial at degrees 0 and 1, becomes four-dimensional at degree 2, and exactly one of the four null vectors is independent of the span of $E_1, \ldots, E_4$. This direction is the canonical minimal-degree E_5.*

The degree-by-degree outcome is recorded in Table 3.

Table 3: Nullity and novelty outcomes in the restricted basis $\{M_1, \ldots, M_5\}$.

Degree d	Cols	Nullity of $\mathbf{M}(d, 5, 95)$	New direction beyond span$(E_1, \ldots, E_4)$?
0	5	0	No
1	10	0	No
2	15	4	**Yes** — E_5 first appears
3	20	8	Yes — E_5 persists

After normalization to primitive integer coefficients and orientation so that the constant M_5 coefficient is positive, the unique canonical E_5 extracted at $d = 2$ is:

$$\begin{aligned} E_5(n) = {} & (-450450 + 675675n - 225225n^2)\, M_1(n) \\ & + (960960n - 120120n^2)\, M_2(n) \\ & + (2534912n - 166016n^2)\, M_3(n) \\ & + (7999488n - 322560n^2)\, M_4(n) \\ & + 258048000\, M_5(n). \end{aligned} \tag{3}$$

By computation, $E_5(n) = 0$ for all 95 tested primes in $[2, 500]$ and $E_5(n) \neq 0$ for all tested composites in $[4, 500]$; no claim is made outside the tested range.

Remark 5.2 (Arithmetic content of the coefficients)**.** *The normalization procedure clears all rational denominators and divides by the integer GCD of numerators. For E_5, the constant M_5 coefficient $258048000 = 2^{11} \times 3^2 \times 5^3 \times 7 \times 127$ is the primitive representative; the remaining coefficients share no non-trivial common factor. The presence of the prime 127 is notable: it does not appear in the analogous coefficients of E_1–E_4, which involve only small primes consistent with the Bernoulli numbers at weights ≤ 10.*

6 Verification and Non-Negativity

6.1 Prime verification

We evaluated $E_5(p)$ for all 95 primes $p \in [2, 500]$ using exact rational arithmetic and confirmed $E_5(p) = 0$ in every case. Representative spot checks:

p	$E_5(p)$
2	0
3	0
5	0
7	0
11	0
97	0
499	0

6.2 Sign pattern at composites

The expressions E_1–E_4 of [1] are non-negative at all composites. The canonical E_5 does *not* share this property.

Proposition 6.1 (Sign distribution)**.** *Among the* 404 *composites in* $[4, 500]$*,* $E_5(n) > 0$ *for* 55 *values,* $E_5(n) < 0$ *for* 349 *values, and* $E_5(n) = 0$ *for none. All* 55 *positive composites are odd; no even composite in* $[4, 500]$ *gives a positive value.*

The structural reason is that E_5 lies in a dimension of the prime-vanishing null space orthogonal to the non-negativity cone spanned by E_1–E_4.

Figure 1 plots $E_5(n)$ at all composites in $[4, 200]$ on a symmetric-logarithmic scale ($\mathrm{sign}(E_5) \cdot \log_{10}(1 + |E_5|)$), which preserves sign information while compressing the 18-order-of-magnitude dynamic range.

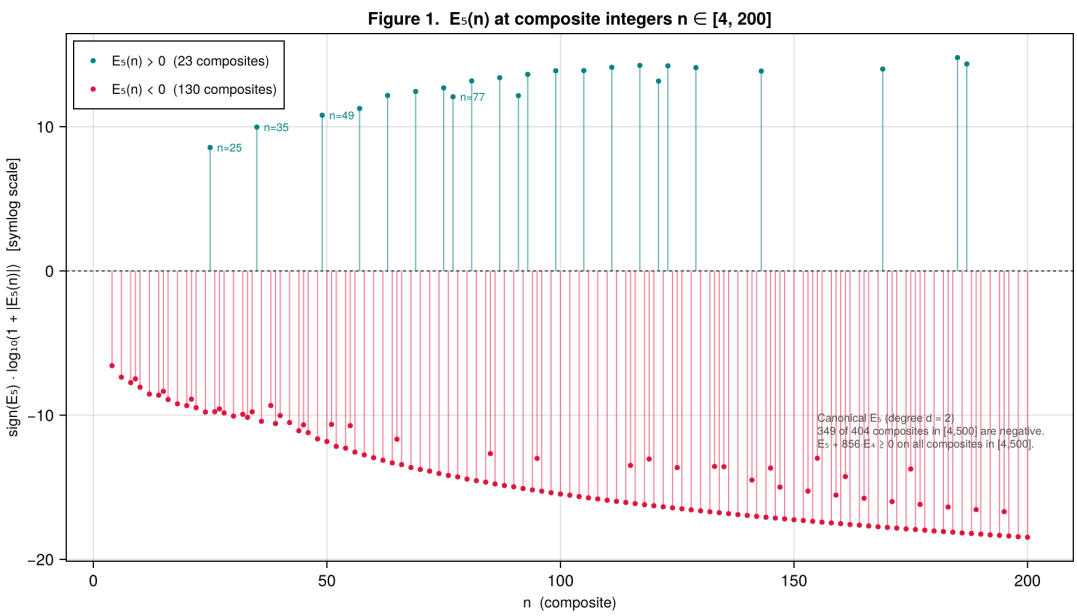

Figure 1: $E_5(n)$ at composite integers in $[4, 200]$ on a symlog scale. Teal stems point upward where $E_5(n) > 0$; crimson stems point downward where $E_5(n) < 0$. All 55 positive composites in $[4, 500]$ are odd; no even composite yields a positive value.

6.3 Non-negative representative

Since $E_4(n) \geq 0$ for all composites and $E_4(p) = 0$ for all primes, the shifted expression $E_5 + kE_4$ preserves the prime-vanishing property for any $k > 0$.

Table 4: Representative evaluations illustrating the sign and scale of E_5.

n	Factorization	$E_5(n)$	Sign
4	2^2	$-3\,693\,690$	negative
9	3^2	$-30\,324\,096$	negative
25	5^2	$363\,888\,000$	positive
35	$5 \cdot 7$	$9\,436\,492\,800$	positive
49	7^2	$62\,655\,312\,384$	positive
100	$2^2 \cdot 5^2$	$-2\,935\,909\,013\,178\,150$	negative

Proposition 6.2 (Smallest non-negative shift)**.** *The smallest positive integer k such that $E_5(n)+kE_4(n) \geq 0$ for all composites in $[4, 500]$ is $k = 856$. The expression E_5+856E_4 achieves its minimum value 4230 at $n = 4$ and spans the same prime-vanishing direction as E_5.*

Table 5: Sign test for E_5 and its non-negative shift at composites in $[4, 500]$.

Expression	Positive values	Negative values	Zero values on tested composites
E_5	55	349	0
$E_5 + 856E_4$	404	0	0

6.4 Closing the conjecture at $a_{\max} = 5$

By appending E_5, the null space of $\mathbf{M}(d, 5, N)$ is fully accounted for. The conjecture of [1] was verified for $a_{\max} = 5$, degrees $d = 2$ through $d = 6$, using $N = 300$ primes:

Specifically, for each degree d the prime-vanishing subspace (the null space of $\mathbf{M}(d, 5, 300)$) was verified to be entirely contained in the $\mathbb{Q}[n]$-span of $\{E_1, \ldots, E_5\}$, i.e. every prime-vanishing expression is a $\mathbb{Q}[n]$-linear combination of the five basis elements.

Table 6: Conjecture verification for $a_{\max} = 5$, $N = 300$ primes. "Prime-van. dim" is the null-space dimension of $\mathbf{M}(d, 5, 300)$; "$\{E_1$–$E_5\}$ span" is the dimension of the $\mathbb{Q}[n]$-span of $\{E_1, \ldots, E_5\}$ evaluated in the ambient $(d+1) \cdot a_{\max}$-dimensional basis space (not the null-space dimension itself); "Holds?" records that the prime-vanishing subspace is contained in that span.

d	Basis dim	Prime-van. dim	$\{E_1$–$E_5\}$ span	Holds?
2	15	4	15	✓
3	20	8	19	✓
4	25	12	23	✓
5	30	16	27	✓
6	35	20	31	✓
7	40	24	35	✓
8	45	28	39	✓

In every case the prime-vanishing subspace is entirely contained in the $\mathbb{Q}[n]$-span of $\{E_1, E_2, E_3, E_4, E_5\}$.

7 Enlarging the Basis and the Appearance of E_6

7.1 The τ-cancellation mechanism

Although M_6 is a pivot column within $\{M_1, \ldots, M_6\}$, the situation changes when M_7 is adjoined. At weight 14, the space $S_{14} = 0$ contains no cusp forms, yet $M_7(n)$ carries both $\tau(n)$ and

$n \cdot \tau(n)$ terms via the $E_2 \cdot \Delta$ component of its depth-1 quasimodular decomposition. Within the $\{M_6, M_7\}$ sub-system evaluated at a prime p, there are therefore two algebraically independent τ-type quantities: $\tau(p)$ and $p \cdot \tau(p)$. A linear combination $\alpha M_6(p) + \beta M_7(p)$ can simultaneously cancel both, leaving a polynomial in p — exactly one free direction in the resulting coefficient space.

Remark 7.1 (M7 alone is also a pivot)**.** *The cancellation requires* both M_6 *and* M_7 *together. Computational verification confirms that* M_7 *is itself a pivot column within* $\{M_1, \ldots, M_7\}$ *when* M_6 *is absent: the nullity of* $\mathbf{M}(d, \{1, 2, 3, 4, 5, 7\}, 95)$ *equals that of* $\mathbf{M}(d, 5, 95)$ *for all tested* $d \in \{2, \ldots, 6\}$*, just as with* M_6 *alone. Only the combined basis* $\{M_1, \ldots, M_7\}$ *admits a null vector, because only then are both* τ*-type terms simultaneously available for cancellation.*

7.2 Nullity gain upon adjoining M_7

Proposition 7.2 (Nullity gains)**.** *When the basis is enlarged from* $\{M_1, \ldots, M_6\}$ *to* $\{M_1, \ldots, M_7\}$*, the exact prime-evaluation matrices exhibit nullity gains* $0, 0, 1, 2, 3$ *at degrees* $d = 2, 3, 4, 5, 6$*, respectively.*

Table 7 records the absolute nullities and gains.

Table 7: Nullity comparison across bases at $N = 95$ primes. The gain first appears at $d = 4$.

d	Nullity $\{M_1$–$M_5\}$	Nullity $\{M_1$–$M_6\}$	Nullity $\{M_1$–$M_7\}$	Gain
2	4	4	4	0
3	8	8	8	0
4	12	12	13	**+1**
5	16	16	18	**+2**
6	20	20	23	**+3**

The gain first appears at $d = 4$. The additional null vectors at $d = 5$ and $d = 6$ are polynomial-degree shifts of the same new direction; no genuinely second expression exists.

7.3 The formula for E_6

The unique new prime-vanishing direction at $d = 4$, normalized to primitive integer coefficients and oriented so that the constant M_7 coefficient is positive, is:

$$\begin{aligned} E_6(n) = (&105367732470 - 277943208789n + 289613157079n^2 \\ &- 145583618619n^3 + 28545937859n^4)\, M_1(n) \\ &+ (-51324522800n^3 + 4292382480n^4)\, M_2(n) \\ &+ (-40313554176n^3 + 1776519808n^4)\, M_3(n) \\ &+ (-32888346624n^3 + 770273280n^4)\, M_4(n) \\ &+ (-7741440000n^3 - 154828800n^4)\, M_5(n) \\ &+ (-483548921856000 + 37196070912000n)\, M_6(n) \\ &+ 892705701888000\, M_7(n). \end{aligned} \tag{4}$$

The same normalization convention (primitive integer coefficients, highest-M_a constant term positive) is used throughout for all E_k. By computation, $E_6(p) = 0$ for all 95 primes in $[2, 500]$ and $E_6(n) \neq 0$ for all 404 composites in $[4, 500]$ (173 positive, 231 negative).

Remark 7.3 (Coefficient magnitudes of E_6)**.** *The largest coefficient in* (4) *is the constant* M_7 *term* $892705701888000 = 2^{10} \times 3^5 \times 5^3 \times 7 \times 691 \times 1303$*. The factor* 691 *is the same Bernoulli*

prime that appears in the denominator of the M_6 tau coefficient c_τ (Appendix A), reflecting the weight-12 modular structure shared by M_6 and the depth-1 component of M_7. The factor 1303 *is the leading coefficient of n^4 in the $c_1(n)$ polynomial of E_4, suggesting a deeper arithmetic relation between E_4 and E_6 that remains to be understood.*

7.4 Conjecture verification with E_6 and extended search to $a_{\max} = 12$

After appending E_6, the extended conjecture was verified for $a_{\max} = 7$, $d = 2$ through $d = 8$, $N = 300$ primes: in each case the prime-vanishing subspace of $\mathbf{M}(d, 7, 300)$ was checked to be entirely contained in the $\mathbb{Q}[n]$-span of $\{E_1, \ldots, E_6\}$.

Table 8: Conjecture verification for $a_{\max} = 7$, $N = 300$ primes. "Prime-van. dim" is the dimension of the prime-vanishing null space of $\mathbf{M}(d, 7, 300)$; "$\{E_1$–$E_6\}$ span dim" is the dimension of the $\mathbb{Q}[n]$-span of $\{E_1, \ldots, E_6\}$ evaluated at 300 primes; "Contained?" records whether the null space is entirely contained in that span.

d	Basis dim	Prime-van. dim	$\{E_1$–$E_6\}$ span dim	Contained?
2	21	4	18	✓
3	28	8	23	✓
4	35	13	28	✓
5	42	18	33	✓
6	49	23	38	✓
7	56	28	43	✓
8	63	33	48	✓

The search was then extended stepwise through $M_8, \ldots, M_{12}$ (weights 16–24). For each new M_a, a closed-form expression was fitted by rational RREF, incorporating all relevant cusp-form types, and the $\{E_1, \ldots, E_6\}$-span test was rerun.

Table 9: Extended search to $a_{\max} = 12$. No new E_i found beyond E_6.

a	Weight $2a$	$\dim S_{2a}$	New cusp type	New E_i found?
8	16	1	$\Delta \cdot E_4$	No
9	18	1	$\Delta \cdot E_6$	No
10	20	1	$\Delta \cdot E_8$	No
11	22	1	$\Delta \cdot E_{10}$	No
12	**24**	**2**	$\Delta \cdot E_{12}$, Δ^2	**No**

Despite large nullity gains at each step (e.g. at $d = 6$: nullity grows from 23 at $a_{\max} = 9$ to 44 at $a_{\max} = 12$), every new null vector is a polynomial-degree shift of an existing E_i. Weight 24 is the first weight at which $\dim S_{2a} = 2$, making M_{12} the most natural candidate for a second τ-cancellation mechanism; that none arises suggests the completeness of $\{E_1, \ldots, E_6\}$ is structural rather than coincidental.

8 Bounded Computational Completeness

Proposition 8.1 (Bounded completeness)**.** *Using exact rational arithmetic, exhaustive searches through $a_{\max} = 12$ and $d \leq 8$ found no independent prime-vanishing directions beyond $E_1, \ldots, E_6$.*

The evidence supporting this proposition is quantitative. Table 10 records the prime-evaluation matrix nullity at $d = 6$ as $a_{\max}$ is incremented from 7 to 12, together with the result of the $\{E_1, \ldots, E_6\}$-span test at each level.

Table 10: Nullity of $\mathbf{M}(6, a_{\max}, 95)$ and span test result as $a_{\max}$ ranges from 7 to 12 ($d = 6$, $N = 95$ primes). "Span test" records whether every prime-vanishing null vector lies in the $\mathbb{Q}[n]$-span of $\{E_1, \ldots, E_6\}$.

$a_{\max}$	Weight $2a_{\max}$	$\dim S_{2a_{\max}}$	Nullity at $d = 6$	New E_i?	Span test
7	14	0	23	Yes (E_6)	✓
8	16	1	30	No	✓
9	18	1	36	No	✓
10	20	1	38	No	✓
11	22	1	41	No	✓
12	**24**	**2**	**44**	**No**	✓

Although the nullity grows substantially at each step (a gain of 21 from $a_{\max} = 7$ to $a_{\max} = 12$), every new null vector at each level is a polynomial-degree shift of an existing E_i; no genuinely new prime-vanishing direction appears. The same span test was verified for $d = 2, \ldots, 8$ at each level with identical conclusions. At weight 24 ($a_{\max} = 12$), the cusp-form space first becomes two-dimensional ($\dim S_{24} = 2$, spanned by $\Delta \cdot E_{12}$ and Δ^2), making M_{12} the most structurally favourable candidate for a second τ-cancellation mechanism. That no new expression arises at this weight strongly suggests that completeness of $\{E_1, \ldots, E_6\}$ is a structural property rather than an artefact of the low-weight regime.

9 Summary of the E_1–E_6 Sequence

The six prime-detecting expressions form a hierarchy constrained by weight and polynomial degree:

Table 11: The E_1–E_6 sequence. The pattern breaks at E_5 due to the weight-12 cusp form barrier, and resumes in a modified form at E_6 via τ-cancellation.

Expr	Min degree d	Highest M_a	Non-neg. on composites?	Notes
E_1	2	M_2	Yes (tested)	Introduces M_2
E_2	3	M_3	Yes (tested)	Introduces M_3
E_3	4	M_4	Yes (tested)	Introduces M_4
E_4	5	M_5	Yes (tested)	Introduces M_5
E_5	**2**	M_5	No (tested)	M_6 excluded; degree resets
E_6	**4**	M_7	No (tested)	τ-cancellation via M_6+M_7

10 Conclusion

This work illuminates two complementary structural phenomena governing prime-vanishing expressions in the MacMahon basis.

The first is the **weight-12 cusp form barrier**: the generating function $U_6(q)$ enters the cusp form space at weight 12, introducing Ramanujan's τ-function, which does not reduce to polynomials in divisor sums. This forces M_6 into the pivot columns of every prime evaluation matrix built from $\{M_1, \ldots, M_6\}$, and compels E_5 to reside in the M_1–M_5 basis at the unexpectedly low degree $d = 2$.

The second is the **τ-cancellation mechanism**: at weight 14, $U_7(q)$ introduces both $\tau(n)$ and $n \cdot \tau(n)$ via its quasimodular $E_2 \cdot \Delta$ component. Within $\{M_1, \ldots, M_7\}$, a linear combination

of M_6 and M_7 simultaneously cancels both τ-type contributions at primes, yielding E_6 at degree $d = 4$.

The extended computational search through $a_{\max} = 12$ and $d \leq 8$ finds no expressions beyond E_6. Within these bounds, no additional independent prime-vanishing directions were found beyond $E_1, \ldots, E_6$ (see Proposition 8.1).

Several directions remain open:

1. **Sign structure.** All 55 positive composites of E_5 in $[4, 500]$ are odd; parity is necessary but not sufficient for positivity. For E_6, 173 positive and 231 negative composites appear. What arithmetic property of n determines the sign?

2. **Theoretical completeness.** A proof that $\{E_1, \ldots, E_6\}$ spans all prime-vanishing expressions for every $a_{\max}$ and d would require understanding why higher cusp-form spaces (e.g. $\dim S_{24} = 2$) do not open new cancellation channels. Recent work on the Eisenstein characterisation of prime-detecting quasimodular forms [4, 5, 6] may provide the theoretical tools for such a proof.

3. **Automorphic framework.** Is there an analogue of the generating functions $U_a(q)$ that organizes the E_k into a coherent automorphic object?

The broader significance of these computations is that they isolate the first obstruction layer and the first recovery layer in partition-theoretic prime detection. In companion work, the weight-12 barrier is proved structurally, explaining why M_6 cannot contribute a new prime-vanishing direction in the restricted basis. A second companion paper then studies the first recovery after adjoining M_7, showing how the cusp sector re-enters through a cancellation mechanism that first becomes visible at degree 4. The present paper therefore serves as the computational foundation for a structural obstruction–recovery theory.

Acknowledgements. The computations were performed in exact rational arithmetic using Julia scripts available at `https://github.com/randsley/PartitionPrimes`. The author thanks colleagues for helpful discussions (names to be supplied).

11 Methods and Reproducibility

All decisive rank, nullity, and basis-comparison computations were carried out over $\mathbb{Q}$ using `Rational{BigInt}` arithmetic in Julia. No floating-point linear algebra was used in any final determination of null-space dimensions. The accompanying repository contains scripts that reproduce every numerical claim: `Paper/extract_paper_data.jl` (E5 extraction and conjecture checks), `Paper/extract_e6_e7.jl` (E6 derivation), `Paper/extract_higher.jl` (search through M_{12}), and `Paper/compute_m6_coefficients.jl` (the P_j polynomials in Appendix A).

A Explicit Polynomials in the Closed-Form of $M_6(n)$

The complete closed-form (2) for $M_6(n)$ requires the six polynomials $P_j(n) \in \mathbb{Q}[n]$ (each multiplying $\sigma_{2j+1}(n)$). These were obtained by solving the 55×22 rational linear system described in Section 3 and verified against direct partition enumeration for $n = 1, \ldots, 30$ with no mismatches.

$$P_0(n) = \frac{550499}{4541644800} - \frac{153617}{371589120}\,n + \frac{2159}{6635520}\,n^2 - \frac{67}{737280}\,n^3 + \frac{11}{1105920}\,n^4 - \frac{1}{2764800}\,n^5$$
$$P_1(n) = \frac{153617}{743178240} - \frac{2159}{8847360}\,n + \frac{67}{819200}\,n^2 - \frac{11}{1105920}\,n^3 + \frac{1}{2580480}\,n^4$$
$$P_2(n) = \frac{2159}{88473600} - \frac{67}{4915200}\,n + \frac{11}{5160960}\,n^2 - \frac{1}{10321920}\,n^3$$
$$P_3(n) = \frac{67}{123863040} - \frac{11}{74317824}\,n + \frac{1}{111476736}\,n^2$$
$$P_4(n) = \frac{11}{3715891200} - \frac{1}{3096576000}\,n$$
$$P_5(n) = \frac{1}{268995133440}$$

The tau coefficient is $c_\tau = -17/150\,450\,048\,000$. The 22 coefficients were extracted as the unique solution to the fitting system (rank 22 over $\mathbb{Q}$).

B Explicit Formula for $E_6(n)$ in Primitive Normal Form

The unique new prime-vanishing direction at $d = 4$ in the basis $\{M_1, \ldots, M_7\}$, normalized to primitive integer coefficients with the constant M_7 coefficient positive, is given by (4) in the main text. For reference, the coefficient polynomials organized by M_a are:

$$c_1(n) = 105367732470 - 277943208789n + 289613157079n^2 - 145583618619n^3 + 28545937859n^4$$
$$c_2(n) = -51324522800n^3 + 4292382480n^4$$
$$c_3(n) = -40313554176n^3 + 1776519808n^4$$
$$c_4(n) = -32888346624n^3 + 770273280n^4$$
$$c_5(n) = -7741440000n^3 - 154828800n^4$$
$$c_6(n) = -483548921856000 + 37196070912000n$$
$$c_7 = 892705701888000$$

so that $E_6(n) = \sum_{a=1}^{7} c_a(n)\, M_a(n)$.

C Expanded Search Logs

Table 12: Degree sweep in the restricted basis $\{M_1, \ldots, M_5\}$, $N = 95$ primes.

Degree d	Cols	Rank	Nullity	Novelty outcome
0	5	5	0	No new direction
1	10	10	0	No new direction
2	15	11	4	E_5 appears
3	20	12	8	E_5 persists
4	25	13	12	E_5 persists
5	30	14	16	E_5 persists

Degree d	Cols	Rank	Nullity	Novelty outcome
6	35	15	20	E_5 persists

Table 13: Nullity comparison after adjoining M_7, $N = 95$ primes.

Degree d	Cols $\{M_1\text{–}M_7\}$	Rank	Nullity gain	Notes
2	21	17	0	No new direction
3	28	20	0	No new direction
4	35	22	1	First appearance of E_6
5	42	24	2	Polynomial-degree shift of E_6
6	49	26	3	Polynomial-degree shift of E_6

D Computational Scaling

The prime evaluation matrix has size $N \times (d+1)a_{\max}$. Rational RREF is the dominant cost; its complexity is $O(N \cdot (d+1)^2 a_{\max}^2)$ arithmetic operations over $\mathbb{Q}$. Measured runtimes on a single core (Julia, `Rational{BigInt}` arithmetic):

Degree d	Matrix size ($N = 300$, $a_{\max} = 5$)	Nullity	RREF time
6	300×35	20	0.16 s
10	300×55	36	0.24 s
20	300×105	76	0.62 s
30	300×155	116	1.13 s

Extensions to $d \leq 50$ are straightforwardly feasible; the bound $d \leq 8$ in the completeness search is conservative and chosen for verification speed.

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E Structure of $M_7(n)$: Two Independent τ-Type Terms

Although the cusp-form space $S_{14} = 0$ (weight 14 has no cusp forms), the generating function $U_7(q)$ is a quasimodular form of depth 1, meaning it involves the quasi-period $E_2(q)$ in its modular transformation law. The depth-1 piece of $U_7(q)$ lives in the weight-12 space $M_{12} = \mathrm{span}\{E_{12}, \Delta\}$, and the $E_2 \cdot \Delta$ component contributes both $\tau(n)$ and $n \cdot \tau(n)$ to the Fourier coefficients of $U_7(q)$. Consequently $M_7(n)$ has the fitted closed form

$$M_7(n) = \sum_{j=0}^{6} Q_j(n)\, \sigma_{2j+1}(n) + c_\tau\, \tau(n) + c_{n\tau}\, n \cdot \tau(n), \tag{5}$$

where $Q_j(n) \in \mathbb{Q}[n]$ are polynomials of degree $\leq 6 - j$, and $c_\tau, c_{n\tau} \in \mathbb{Q}$ with $c_{n\tau} \neq 0$. The 30 rational coefficients (28 divisor-sum terms plus c_τ and $c_{n\tau}$) are determined uniquely by fitting a 70×30 rational linear system against $M_{\mathrm{direct}}(7, n)$ for $n = 1, \ldots, 70$; the system has rank 30 and the solution is verified to reproduce $M_7(n)$ for $n = 1, \ldots, 30$ with no mismatches.

Why two terms are needed for τ-cancellation. At a prime p, the quantities $\tau(p)$ and $p \cdot \tau(p)$ are expected to be independent of divisor sums and of each other; this is consistent with standard conjectures on the Ramanujan tau function. Numerically, no linear dependence was detected among these quantities across all primes tested ($p \leq 500$). Within the $\{M_6, M_7\}$ sub-system evaluated at p, there are therefore *two* independent τ-type quantities. Cancelling both simultaneously requires exactly two free parameters, i.e. at least two basis functions each carrying a τ-type term. Since M_6 carries $\tau(p)$ (one term) and M_7 carries both $\tau(p)$ and $p \cdot \tau(p)$ (two terms), a suitable linear combination $\alpha M_6(p) + \beta M_7(p)$ can annihilate both simultaneously

— leaving a pure polynomial in p and creating the one-dimensional new null-space direction that is E_6. This two-term structure of M_7 is therefore not incidental but the precise algebraic condition that makes E_6 possible.